DSC 140A - Homework 07

Homework 07 Instructions. You may collaborate with whomever, including AI. Each student must submit their own homework even if they worked with others. We will not be grading for correctness, but we will be grading for completeness. Please turn out a single PDF file that contains your solutions to all of the problems. We recommend using LaTeXwith our template. You can use markdown but make sure each problem is clearly labeled. For coding problems, we recommend using Jupyter notebooks, covert them into PDF, and then merge the PDF with your math problems solutions generated in LaTeX.

Problem 1.

Let X be a continuous random variable, and let Y be a binary random variable. Suppose the class conditional densities are know to be:

$$p_X(x \mid Y = 0) = \begin{cases} 1/5, & \text{if } 0 \le x \le 2 \\ 1/3, & \text{if } 2 < x \le 3 \\ 1/15, & \text{if } 3 < x \le 7 \end{cases}$$

$$p_X(x \mid Y = 1) = \begin{cases} 1/6, & \text{if } 0 \le x \le 1 \\ 1/8, & \text{if } 1 < x \le 5 \\ 1/6, & \text{if } 5 < x \le 7 \end{cases}$$

Suppose also that $\mathbb{P}(Y = 0) = 0.4$ and $\mathbb{P}(Y = 1) = 0.6$.

For what values of $x \in [0, 7]$ will the Bayes classifier predict $y = 1$?

Problem 2.

Let X be a continuous random variable, and let Y be a random class label (1 or 0). Recall that the Gaussian probability density function (pdf) is given by:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Suppose $p_X(x \mid Y = 1)$ is a Gaussian pdf with $\mu = 2$ and $\sigma = 3$ and that $p_X(x \mid Y = 0)$ is also Gaussian with $\mu = 5$ and $\sigma = 3$. Suppose, too, that $\mathbb{P}(Y = 1) = \mathbb{P}(Y = 0) = \frac{1}{2}$.

Recall that the Bayes error is the probability that the Bayes classifier makes an incorrect prediction. What is the Bayes error for this distribution? Show your work.

Hint: you'll want some way to compute the area under a Gaussian. You can use the tables that appear in the back of your statistics book, or you can use something like `scipy.stats.norm.cdf`. We'll let you read the documentation to see how to use it, but it may be helpful to remember that if F is the cumulative density function for a distribution with density f, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

Problem 3.

The file linked below contains a data set of 150 samples. The first column contains a single continuous feature, X, assumed to have been drawn from an unknown probability density. The second column contains the binary class label Y.

`https://f000.backblazeb2.com/file/jeldridge-data/011-univariate_density_estimation/data.csv`

In this problem, use a histogram estimator with bins $[0, 1.5), [1.5, 3), \ldots, [13.5, 15)$ to estimate the requested probabilities. In each part, show your code and provide your reasoning.

Hint: you may find `np.histogram` useful.

a) Estimate $\mathbb{P}(Y = 1 \,|\, X = 6.271)$ directly.

b) Estimate $\mathbb{P}(Y = 1 \,|\, X = 6.271)$ by estimating all of: 1) the marginal density $p_X(x)$, 2) the class-conditional density $p_X(x \,|\, Y = 1)$, and 3) the class prior $\mathbb{P}(Y = 1)$, and then applying Bayes' rule.

c) For what values of $x \in [0, 15]$ will the Bayes classifier predict $y = 1$?

Problem 4.

The file linked below contains a data set of 150 samples. The first column contains a single continuous feature, X, assumed to have been drawn from an unknown probability density. The second column contains the binary class label Y.

`https://f000.backblazeb2.com/file/jeldridge-data/011-univariate_density_estimation/data.csv`

In the first problem on this homework, you used a histogram estimator to apply the Bayes classification rule. In this problem, you will instead estimate the class-conditional densities by fitting Gaussians using the method of maximum likelihood. In each part, show your code and provide your reasoning.

Note: you should *not* use `sklearn`, `scipy`, or any other library to fit the Gaussian densities. You should instead calculate the maximum likelihood estimates directly (using `numpy` or `pandas` to do this is fine).

a) Estimate the class-conditional densities $p_X(x \,|\, Y = 0)$ and $p_X(x \,|\, Y = 1)$ with Gaussians using the method of maximum likelihood and report the estimated parameters.

b) Let $\tilde{p}_X(x \,|\, Y = 0)$ and $\tilde{p}_X(x \,|\, Y = 1)$ be the estimated class-conditional densities from the previous part, and let $\tilde{\mathbb{P}}(Y = 1)$ be the estimate for $\mathbb{P}(Y = 1)$.

Plot $\tilde{p}_X(x \,|\, Y = 0) \cdot \tilde{\mathbb{P}}(Y = 0)$ and $\tilde{p}_X(x \,|\, Y = 1) \cdot \tilde{\mathbb{P}}(Y = 1)$ on the same axis. Label your plot so that the grader can tell which Gaussian corresponds to which class. Remember to show your code.

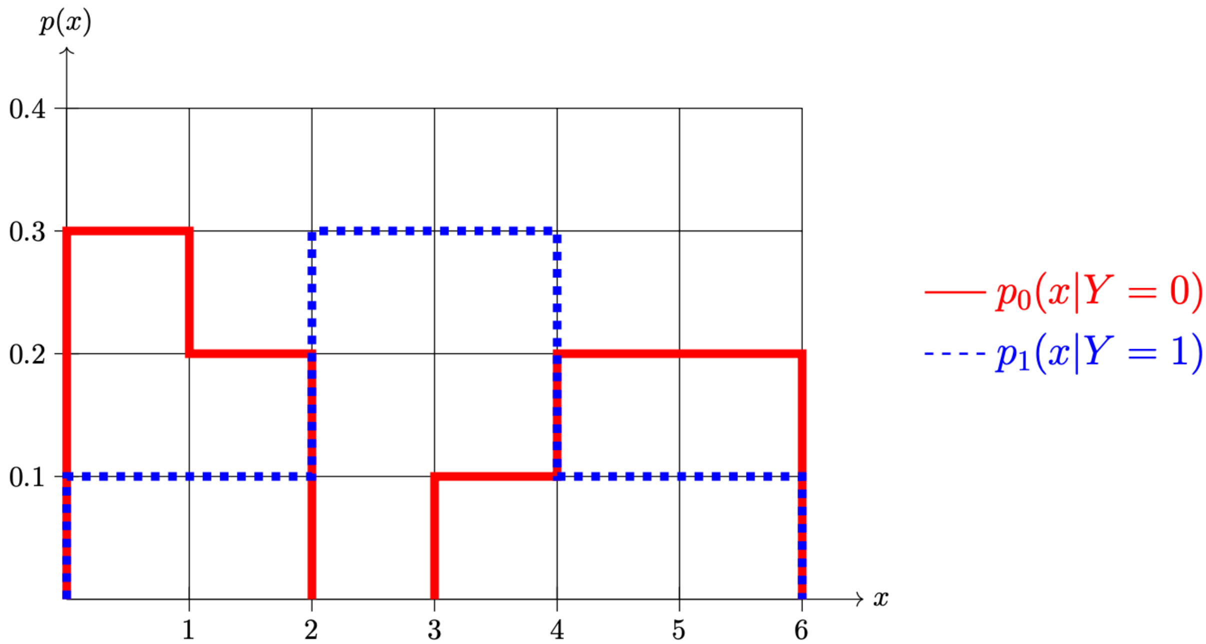
c) Suppose a new point is observed at $x = 6.271$. What class does the Bayes classifier predict for x when the estimated densities and probabilities are used? Remember to show your code, and determine the predicted class through calculation (and not by visual inspection of the plot from the previous part).

d) The `scipy.optimize.fsolve` function can be used to find the *roots* of a function f; that is, the places where function $f(x) = 0$. Use `fsolve` to find the decision boundary for the classifier you trained above. Show your code.

Note that there may actually be multiple decision boundaries at the extremes, but there is one clear decision boundary in the middle of the data, as should be evident in your plot; find that one.

Problem 5.

Shown below are two conditional densities, $p_1(x \,|\, Y = 1)$ and $p_0(x \,|\, Y = 0)$, describing the distribution of a continuous random variable X for two classes: $Y = 0$ (the solid line) and $Y = 1$ (the dashed line). You may assume that both densities are piecewise constant.



a) Suppose that $\mathbb{P}(Y=1) = \mathbb{P}(Y=0) = 0.5$. What is $\mathbb{P}(1 \le X \le 3)$?

b) Suppose $\mathbb{P}(Y=1) = \mathbb{P}(Y=0) = 0.5$. What is the prediction of the Bayes classifier at $x = 1.5$?

c) Suppose again that $\mathbb{P}(Y=1) = \mathbb{P}(Y=0) = 0.5$. What is the Bayes error with respect to this distribution?

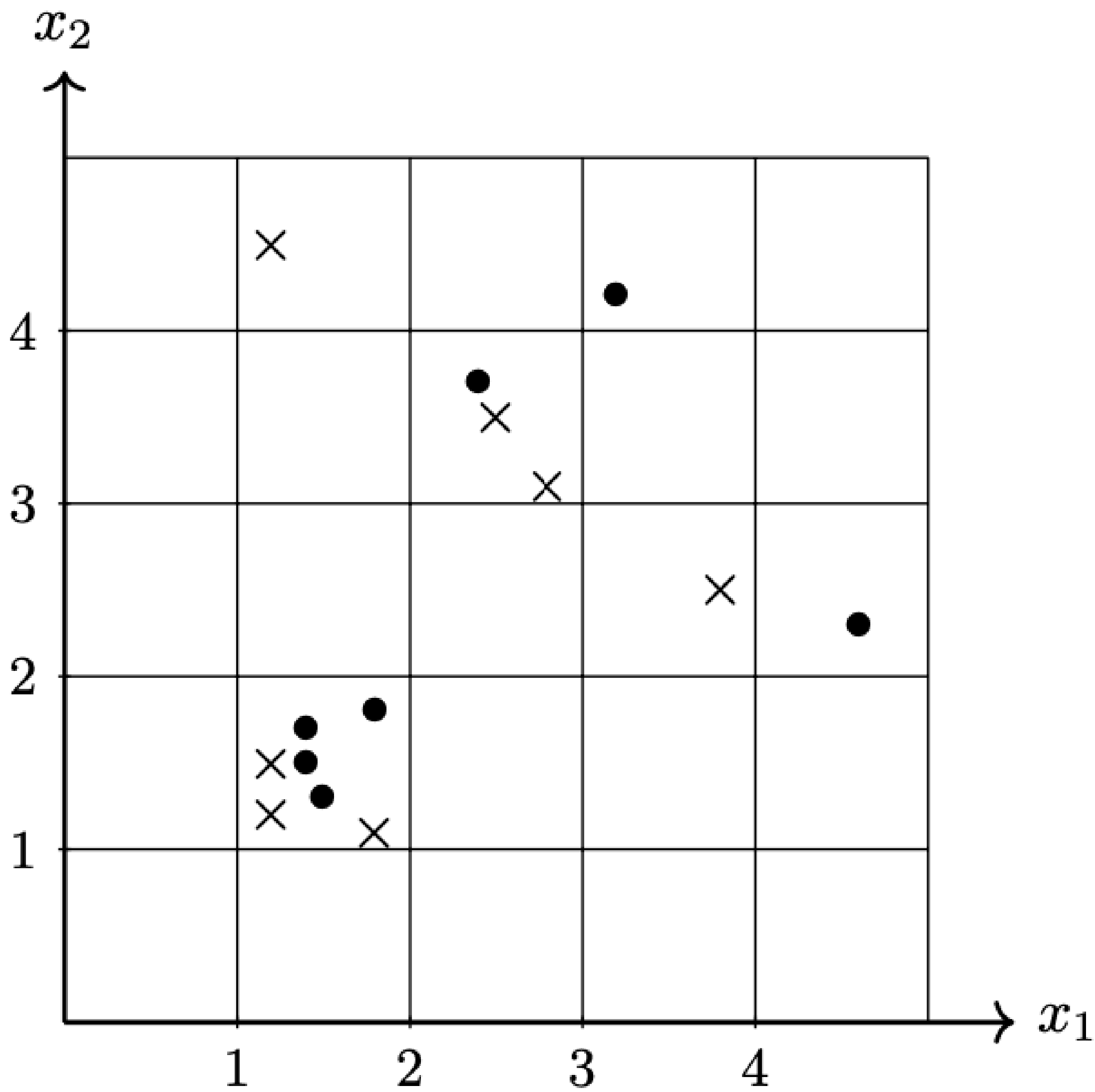
d) Now suppose $\mathbb{P}(Y=1) = 0.7$ and $\mathbb{P}(Y=0) = 0.3$. What is $\mathbb{P}(1 \le X \le 3)$?

e) Suppose again that $\mathbb{P}(Y=1) = 0.7$ and $\mathbb{P}(Y=0) = 0.3$. What is the prediction of the Bayes classifier at $x = 1.5$?

f) Suppose again that $\mathbb{P}(Y=1) = 0.7$ and $\mathbb{P}(Y=0) = 0.3$. What is the Bayes error with respect to this distribution?

Problem 6.

Consider the following data set of 14 points in $\mathbb{R}^2$. Each point has a label in $\{1, -1\}$. Points from Class 1 are marked with $\times$ and points from Class -1 are marked with $\bullet$.



Suppose the class-conditional density estimates are computed from this data using a histogram density estimator using the 1×1 bins shown in the figure above.

If these estimates are used in place of the true class-conditional densities in the Bayes classifier, what will be the training error of the classifier? That is, what percentage of the data above will be misclassified? You may leave your answer as a fraction or a decimal.